

EXPERIMENT – 5b

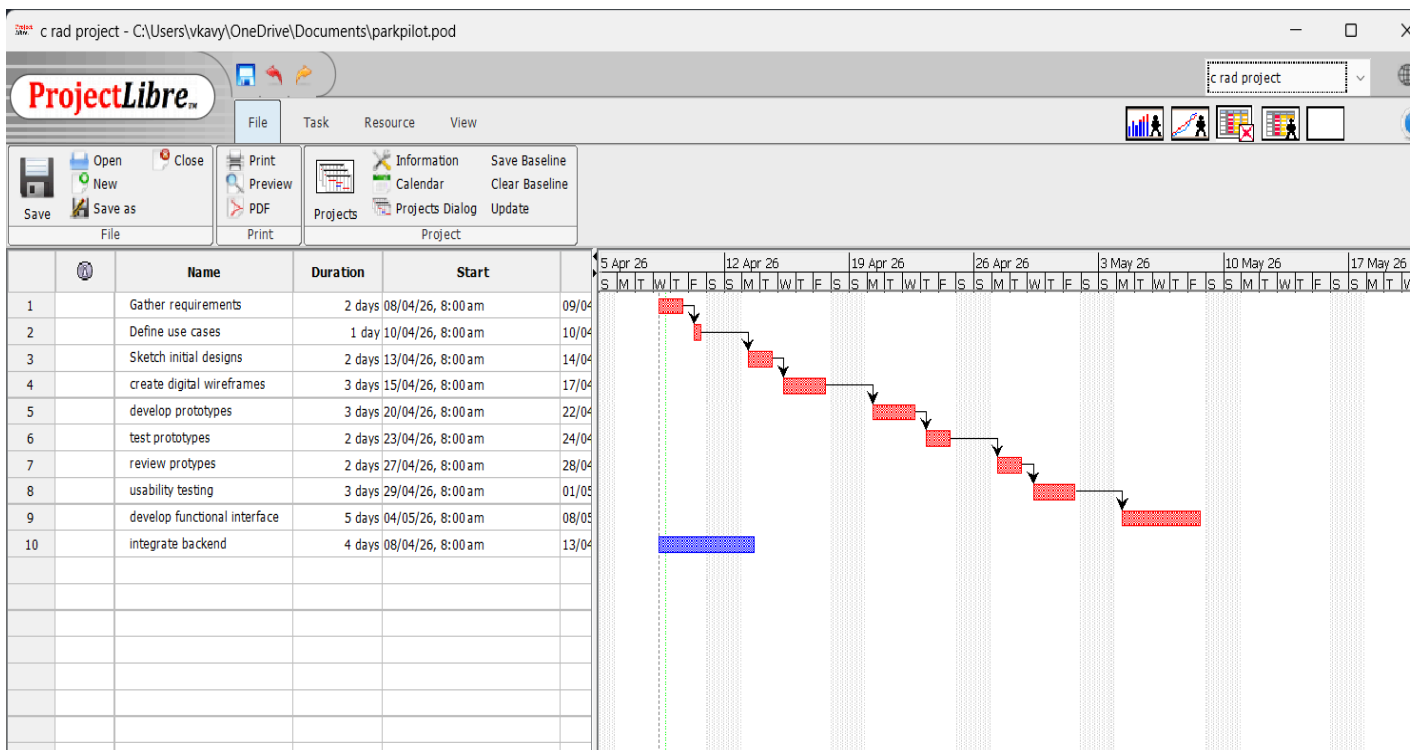
RAD Model UI Lifecycle (Using OpenProj)

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The aim is to simulate the lifecycle stages of UI design using the Rapid Application Development (RAD model) and develop a small interactive interface for the ParkPilot using OpenProj.



ParkPilot – RAD Model UI Lifecycle (Using OpenProj)

Step 1: Requirements Planning (ParkPilot)

In this phase, the core functionality of the ParkPilot application is identified.

Features identified:

- Find nearby parking slots
- Live slot availability
- Booking system
- Navigation to parking location
- User login and signup

Use Cases:

- User logs in and views nearby parking slots
- User selects a slot and completes booking
- New user registers and accesses the system

This stage establishes the core idea of ParkPilot as a smart parking solution with real-time capabilities.

Step 2: User Design

This phase focuses on designing the interface of ParkPilot.

Design Activities:

- Create sketches for:
- Home screen (map view)
- Slot selection screen
- Booking confirmation screen
- Login and signup screens

Wireframes:

- Login: Username, Password, Login button
- Register: Username, Email, Password, Confirm Password
- Home: Map with available parking slots
- Slot Page: Slot details with booking option

This stage includes both familiar navigation elements (buttons, forms) and innovative features (smart slot suggestions).

Step 3: Rapid Prototyping

In this stage, the wireframes are converted into an interactive prototype.

Activities:

- Develop a clickable prototype using prototyping tools
- Implement navigation flow:

Login → Home → Map → Slot → Booking

Testing:

- Share prototype with users
- Collect feedback on usability and navigation

Output:

- Interactive prototype of ParkPilot

This phase validates the usability of the ParkPilot flow.

Step 4: User Acceptance / Testing

This phase evaluates the prototype through user feedback.

Activities:

- Conduct user and stakeholder reviews
- Perform usability testing:
- Check ease of booking
- Evaluate map clarity
- Identify navigation issues

Output:

- Documented feedback and usability results

This stage ensures the application is user-friendly and efficient.

Step 5: Implementation

In this phase, the final application is developed.

Activities:

- Develop frontend interface:
- Map interface
- Booking system
- Integrate backend:
- User authentication
- Parking slot database
- Real-time updates

Output:

- Fully functional ParkPilot application

This phase transforms the prototype into a working product.

FINAL FLOW (RAD in ParkPilot)

Requirements → Design → Prototype → Test → Implementation

The RAD model enables rapid development, continuous feedback, and user-centered design, making it suitable for applications like ParkPilot.